COURSE PROJECT PROPOSAL FOR MICROCONTROLLER AND IoT

Automatic Lighting System and Light Intensity Monitoring Based on Light Sensors and IoT

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CHAPTER I

INTRODUCTION

1. Background

Nowadays, electricity has become an essential need in daily life. Almost all human activities rely on electrical energy, ranging from industrial and educational sectors to public facilities such as streetlights and garden lights. However, in reality, public lighting facilities often do not function optimally. Some lights do not turn on at night due to damage or lack of maintenance, while in other places, lights are left on continuously even though it is already bright.

These issues cause energy waste and reduce the efficiency of power usage. In addition, systems that are still manually controlled make lighting management less practical. Therefore, an automated system is needed to control the turning on and off of lights automatically based on the light intensity in the surrounding environment.

With the development of the Automatic Lighting System and Light Intensity Monitoring Based on Light Sensor and IoT, it is expected that electrical energy usage can become more efficient. The system allows lights to turn on when the environment is dark and off when it is bright automatically, and it can be monitored remotely via the Internet of Things (IoT) platform. Besides facilitating supervision, the system also supports energy-saving efforts and more effective maintenance of public facilities.

2. Problem Statement

Based on the background described, the problem statements of this project are as follows:

1. How to design an automatic lighting system that can turn lights on and off automatically based on a light sensor?
2. How to add a remote-based manual control feature to the automatic lighting system?
3. How to monitor the light intensity level in real-time using Internet of Things (IoT) technology?

4. How can this system improve energy efficiency through automatic control, as well as time efficiency through remote lamp control?

3. Project Objective

The objectives of developing the Automatic Lighting System and Light Intensity Monitoring Based on Light Sensor and IoT are as follows:

1. To design and build an automatic lighting system that can turn lights on and off automatically based on the surrounding light intensity.
2. To add a remote-based manual control feature to increase the flexibility and ease of system operation.
3. To integrate the system with IoT technology so that light intensity and lamp status can be monitored in real-time through an online platform.
4. To improve energy efficiency and time efficiency in managing and monitoring the lighting system.

4. Project benefits

This project is expected to provide benefits in the application of microcontrollers and Internet of Things (IoT) technology as an innovative solution in automation systems. Through this project, it can be demonstrated that the use of microcontrollers and IoT not only facilitates the process of controlling lighting systems but also plays an important role in improving energy efficiency and time efficiency. Furthermore, this project can serve as a basis for the development of similar automation systems in the future, focusing on energy savings and increased operational effectiveness.

CHAPTER II
LITERATURE REVIEW

1. Microcontroller

   A microcontroller (MCU) is an integrated circuit designed to control specific functions within an embedded system. A microcontroller usually consists of a central processing unit (CPU), memory, and input/output (I/O) ports, all integrated into a single chip. Unlike microprocessors used in general-purpose computers, microcontrollers are intended to control specific functions within a larger device.

   In its operation, a microcontroller receives data from sensors or input devices, temporarily stores it in internal memory, processes the data according to embedded program instructions, and outputs the results through I/O ports or actuators. Microcontrollers are designed to be cost- and energy-efficient since all components such as RAM, ROM/Flash, I/O ports, timers, and peripherals are integrated into the chip.

   Due to their efficient and flexible capabilities, microcontrollers are widely used in various applications such as vehicles, household appliances, industrial machines, robotics, and Internet of Things (IoT) devices. In other words, microcontrollers act as the "brain" of many modern automated systems used in daily life.

2. Sensors

   A sensor is a device that detects changes in physical quantities such as pressure, force, electricity, light, motion, humidity, temperature, speed, or other environmental phenomena. When a change occurs, the device converts the detected input into an output that can be understood by humans. This output can be received through the device itself or sent electronically through a network to be displayed or processed into useful information.

   The sensor operation process starts with detecting environmental parameters, such as temperature, pressure, light, humidity, motion, or chemicals. Changes detected by the sensor are converted into electrical signals via transducer components. These electrical signals are usually weak and need amplification before further analysis. After amplification, the signals often undergo conditioning to improve quality or reduce interference.

In many applications, the analog signal generated is converted into digital data using an Analog-to-Digital Converter (ADC). This digital data is processed by a control system or computer to produce useful information or control other devices. The final output of sensor data processing can be displayed to users or used to automate certain actions, such as activating alarms, controlling temperature, or monitoring environmental conditions in real-time.

Some types of sensors used in various applications include:

Berikut ini merupakan beberapa jenis sensor dalam berbagai aplikasi:

1) Proximity
2) Temperature
3) Ultrasonic
4) Radiation
5) Smoke
6) Light
7) Imaging
8) Pressure
9) Humidity

3. IoT

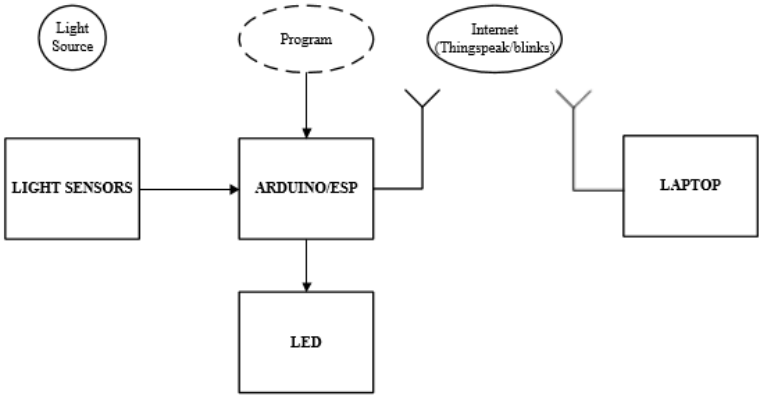
The Internet of Things (IoT) is a concept where physical devices are interconnected via the internet to exchange data automatically without direct human interaction. These devices are equipped with sensors, actuators, processors, and internet connectivity to collect, process, and transmit data to other systems or cloud platforms. IoT consists of three main components: hardware (such as microcontrollers and sensors), communication networks (Wi-Fi/Bluetooth), and IoT platforms (such as Blynk or ThingSpeak) to display and analyze data.

IoT provides significant benefits such as energy efficiency, automation, and increased comfort and security. This technology forms the basis of smart systems such as smart homes, smart cities, and smart industries, allowing real-time monitoring and control of activities via smartphones or computers.

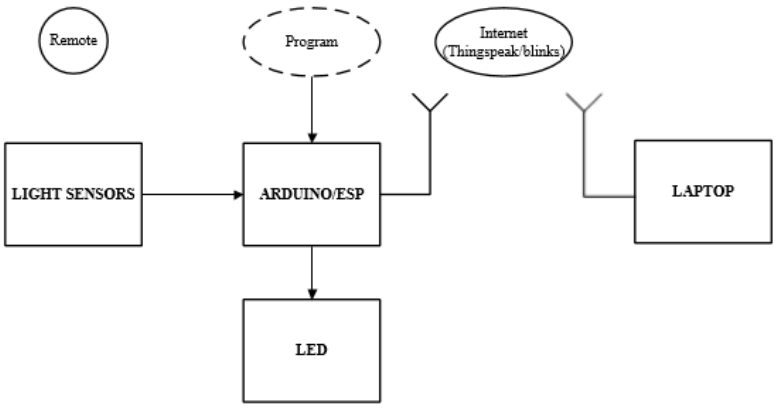
CHAPTER III
DESIGN METHODOLOGY

1. System Block Diagram

   Automatic Mode (Light Intensity):

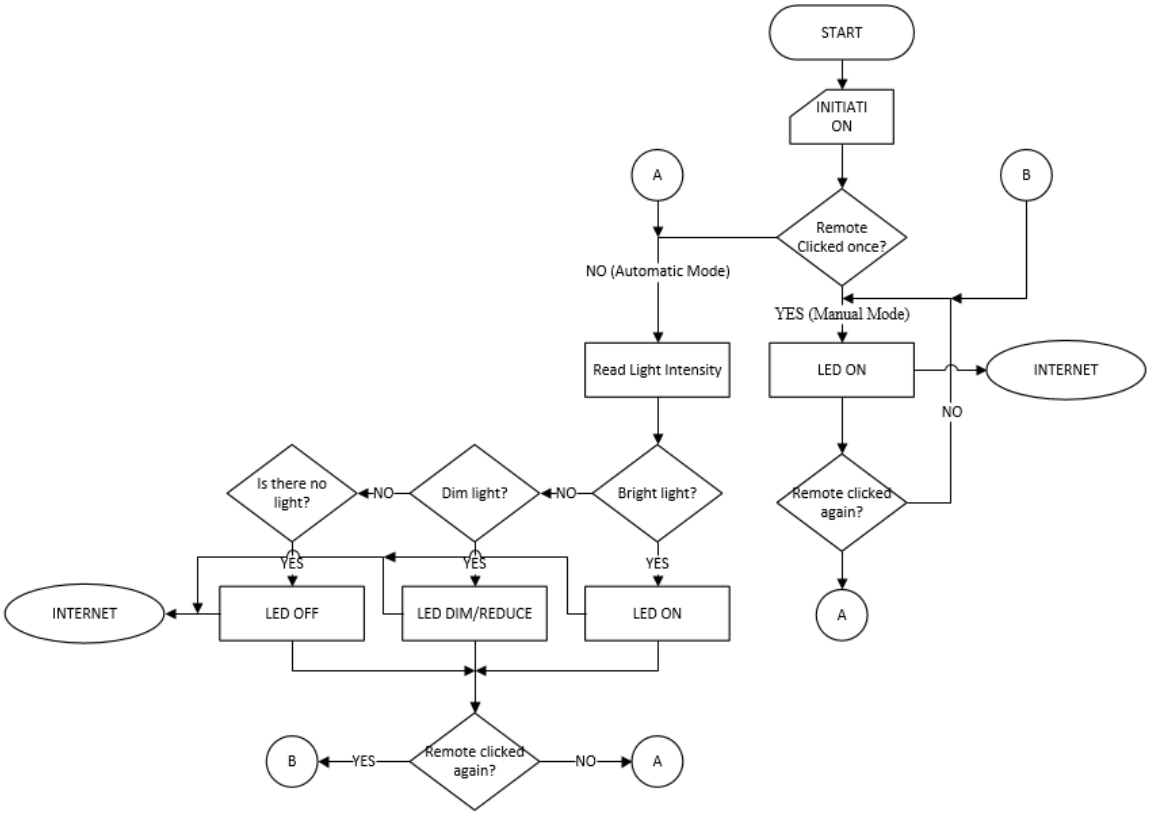


   Manual Mode (Infrared Remote):



   The system operates in two modes: automatic mode and manual mode. In automatic mode, a light sensor in the form of a phototransistor is used to detect the surrounding light intensity, converting the light signal into an electrical signal processed by the Arduino microcontroller to turn the lights on or off automatically. Meanwhile, in manual mode, users can control the lights remotely via an IoT connection using a remote, allowing the system not only to save electrical energy but also to provide time efficiency and ease of operation.

2. Flowchart



3. Testing Procedure

The system development process begins by designing the circuit according to the block diagram, followed by programming the microcontroller using the Arduino IDE to manage the sensor operation and IoT connection. Once the device is assembled, testing is conducted to ensure that the light sensor can accurately detect light intensity, and that the lights can turn on and off properly both automatically and through remote control.

CHAPTER 4 IV
SCHEDULE AND BUDGET

1. Project Schedule

Project Schedule	
Date	**Activity Description**
22 – 10 – 2025	Submission of project proposal, review, and group discussion
29 – 10 – 2025	Project implementation and group discussion
5 – 11 – 2025	Project implementation and group discussion
12 – 11 – 2025	project implementation and group discussion
19 – 11 – 2025	Final presentation, project demo, and submission of report

2. Budget Plan

No.	Item Description	Quantity	Unit Price (Rp)	Subtotal (Rp)
1	ESP8266 devkit	1	28.900	28.900
2	Phototransistor (Receiver)	1	4.500	4.500
3	Red LED	1	2.400	2.400
4	USB Port data type - C	1	4.900	4.900
5	Breadboard / Project board	1	15.000	15.000
6	Infrared LED (Receiver)	1	4.500	4.500
7	Infrared Remote (Transmitter)	1	15.000	15.000
8	Kabel, konektor	1	10.000	10.000
TOTAL ESTIMATED			**85.200**	

CHAPTER V
CONCLUSION

1. Summary

   From the design of this automatic lighting system, it can be concluded that the utilization of microcontrollers and IoT technology is capable of improving both energy and time efficiency. The system can automatically turn lights on or off based on the surrounding light intensity, while also allowing users to control the lights manually through an application or remote. Thus, the system not only supports energy savings but also provides convenience in monitoring and controlling lighting in real-time.